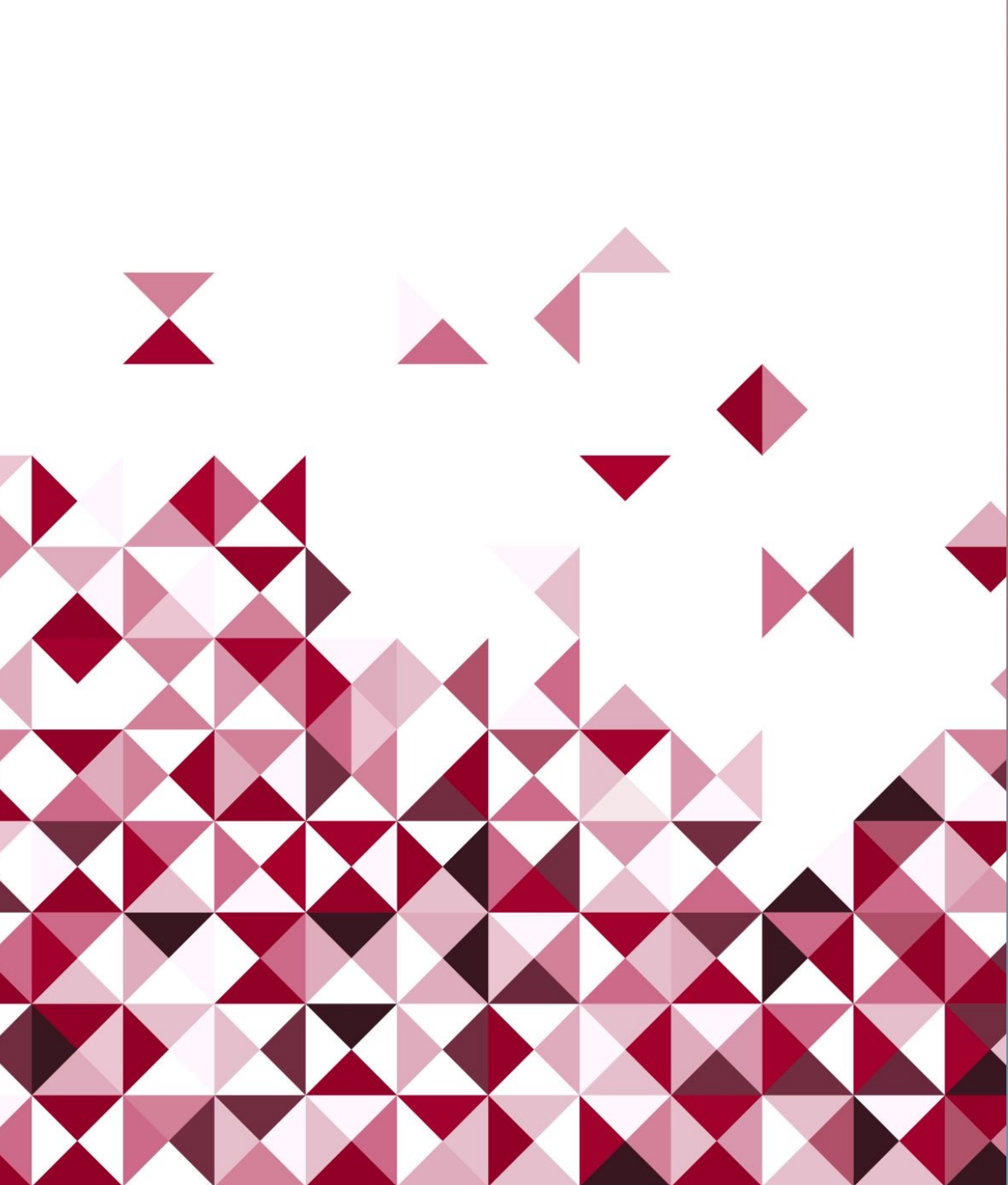




Stat 3113: Lesson 13

The Poisson Probability
Distribution

Definition:

- A random variable X is said to have a **Poisson distribution** with parameter μ ($\mu > 0$) if the pmf of X is:

$$p(x) = \begin{cases} \frac{e^{-\mu} \mu^x}{x!}, & x = 0, 1, 2, 3, \dots \\ 0, & \text{otherwise} \end{cases}$$

➤ Notes:

1. The parameter μ is often a *rate per unit time or area*.
2. The Poisson Distribution is often used in *queuing theory*.
3. Two possible notations: $X \sim \text{Poisson}(\mu)$, or $X \uparrow \sim \text{POI}(\mu)$.

➤ Expected Value and Variance:

$$E(X) = V(X) = \mu$$

Queuing theory, also known as queueing theory, is a branch of mathematics and operations research that focuses on the study of queues or waiting lines.

➤ In Excel:

$$P(X = x) \rightarrow = \text{poisson.dist}(x, \mu, \text{false})$$

$$P(X \leq x) \rightarrow = \text{poisson.dist}(x, \mu, \text{true})$$

NOTE:

The Poisson probability mass function is very close to the binomial probability mass function when n is large, p is small, and $\lambda = np$.

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TABLE 4.1 An example of the Poisson approximation to the binomial probability mass function*

x	$P(X = x),$ $X \sim \text{Bin}(10,000, 0.0002)$	$P(Y = x),$ $Y \sim \text{Bin}(5000, 0.0004)$	Poisson Approximation, Poisson (2) $\rightarrow np = \lambda$
0	0.135308215	0.135281146	0.135335283
1	0.270670565	0.270670559	0.270670566
2	0.270697637	0.270724715	0.270670566
3	0.180465092	0.180483143	0.180447044
4	0.090223521	0.090223516	0.090223522
5	0.036082189	0.036074965	0.036089409
6	0.012023787	0.012017770	0.012029803
7	0.003433993	0.003430901	0.003437087
8	0.000858069	0.000856867	0.000859272
9	0.000190568	0.000190186	0.000190949

*When n is large and p is small, the $\text{Bin}(n, p)$ probability mass function is well approximated by the Poisson (λ) probability mass function (Equation 4.9), with $\lambda = np$. Here $X \sim \text{Bin}(10,000, 0.0002)$ and $Y \sim \text{Bin}(5000, 0.0004)$, so $\lambda = np = 2$, and the Poisson approximation is Poisson(2).

Poisson Process

- The **Poisson process** is one of the most widely-used counting processes.
$$N(t) \sim \text{POI}(\lambda t) \quad (\text{Homogeneous Poisson Process})$$
- It is usually used in scenarios where we are counting the occurrences of certain events that appear to happen at a certain rate, but completely at random (without a certain structure).
- For example, suppose that from historical data, we know that earthquakes occur in a certain area with a rate of 2 per month. Other than this information, the timings of earthquakes seem to be completely random. Thus, we conclude that the Poisson process might be a good model for earthquakes.

Credit: https://www.probabilitycourse.com/chapter11/11_1_2_basic_concepts_of_the_poisson_process.php

Poisson Process

In practice, the Poisson process or its extensions have been used to model.

- ✓ the number of car accidents at a site or in an area;
- ✓ the location of users in a wireless network;
- ✓ the requests for individual documents on a web server;
- ✓ the outbreak of wars;
- ✓ photons landing on a photodiode.

Example 1: Suppose the number X of tornadoes observed in a particular region during a 1-year period has a Poisson distribution with $\mu = 8$.

- a. What is the probability the number of tornados is exactly 5 during a 1-year period?

$$X \sim \text{Poi}(\mu = 8)$$
$$P(X=5) = \frac{e^{-8} 8^5}{5!} = \text{poisson.dist}(5, 8, \text{false}) \approx 0.0816.$$

- b. What is the probability the number of tornados is at most 5?

$$\begin{aligned} P(X \leq 5) &= F(5) \\ &= \sum_{x=0}^5 \frac{e^{-8} 8^x}{x!} \\ &= \text{poisson}(5, 8, \text{true}) \\ &\approx 0.1912 \end{aligned}$$

c. What is the probability between 6 and 9 (inclusive) tornados are observed?

$$\begin{aligned} P(6 \leq X \leq 9) &= \sum_{x=6}^9 \frac{e^{-8} 8^x}{x!} \\ &= \sum_{x=6}^9 \text{poisson.dist}(x, 8, \text{false}) \\ &\approx 0.5254. \end{aligned}$$

$$\begin{aligned} P(6 \leq X \leq 9) &= P(X \leq 9) - P(X \leq 5) \\ &= F(9) - F(5) \\ &= \text{poisson.dist}(9, 8, \text{true}) - \text{poisson.dist}(5, 8, \text{true}) \\ &\approx 0.5254. \end{aligned}$$

d. What is the probability more than 10 tornados are observed?

$$\begin{aligned} P(X > 10) &= 1 - P(X \leq 10) \\ &= 1 - F(10) \\ &= 1 - \text{poisson.dist}(10, 8, \text{true}) \approx 0.1841 \end{aligned}$$

e. What is the mean number of tornadoes during a 1-year period, and what is the standard deviation of the number of observed tornadoes?

$$\bullet E[X] = \mu = 8$$

$$\bullet \text{Var}(X) = \mu = 8 \Rightarrow \sigma_X = \sqrt{8} = 2\sqrt{2} \approx 2.8284 \approx 3 \text{ tornadoes.}$$

Example 2: Suppose that the number of drivers who travel between a particular origin and destination during a designated time period is a Poisson Process with a mean number of drivers of 10 per hour.

$$N(t) \sim \text{POI}(10t)$$

a) What is the probability that the number of drivers will be at most 3?

$$\begin{aligned} IP(N(t) \leq 3) &= F(3) = \text{poisson.dist}(3, 10, \text{true}) \\ &\approx 0.0103. \end{aligned}$$

b) What is the probability that the number of drivers will exceed 7?

$$\begin{aligned} IP(N(t) > 7) &= 1 - IP(N(t) \leq 7) \\ &= 1 - F(7) = \text{poisson.dist}(7, 10, \text{true}) \approx 0.7798 \end{aligned}$$

c) What is the probability that the number of drivers will be between 5 and 8 (non-inclusive)?

$$\begin{aligned} IP(5 < N(t) < 8) &= p(6) + p(7) \longrightarrow \text{poisson.dist}(x, 10, \text{false}) \\ IP(5 < N(t) < 8) &\stackrel{a)}{=} F(7) - F(5) \longrightarrow \text{poisson.dist}(x, 10, \text{true}) \\ &\approx 0.1531 \end{aligned}$$

Reading

Chapter 4 – Sections 4.1, 4.2

Next Topic



Continuous Random Variables



Chapter 3 - **Section 3.3**